Classifier-Causal Mask Discovery with Lexicographic Trust-Region Guidance for Localized Diffusion Counterfactuals

Anonymous Author(s)

Abstract

Visual counterfactuals must answer two connected questions: where may an image change to affect an explained classifier, and how can that change be realized without unnecessary collateral variation? Saliency alone answers the first question only correlationally, while permanently weighted diffusion guidance does not reliably prioritize target success, identity, locality, and non-target preservation. We introduce a training-free framework that first verifies semantic regions through paired, same-seed diffusion interventions and freezes their effects in a classifier-specific influence graph. For a new source, the graph and Grad-CAM++ evidence select a minimal verified semantic support before generation. Inside that support, adaptive lexicographic trust-region causal concept intervention (Adaptive TR-CCI) measures a predicted-clean decode, requests attainable target progress, minimizes the best achievable identity/locality violation, and then minimizes non-target drift. A final latent restoration stage applies the same priority order without additional U-Net or scheduler transitions. In a completed fixed-mask study of 200 unique paired smile-removal sources, adaptive trust-region CCI reduces non-target drift by 11.8% relative to Fixed-weight CCI (0.0635 versus 0.0720) at a one-percentage-point lower Target@0.8 rate (37.5% versus 38.5%). Identity and outside-mask error remain close, at additional runtime. Automatic mask-discovery and two 300-image attacked evaluations are specified but remain pending. The causal claim concerns controlled interventions within the frozen classifier–generator system, not causation in the physical world.

Keywords: visual counterfactuals, causal intervention, semantic mask discovery, diffusion guidance, trust-region optimization

1 Introduction

A visual counterfactual explains a classifier by showing how an input could be changed to obtain a desired output. For faces, crossing the target boundary is not enough: the result should preserve the person's identity, avoid changing unrelated predicted attributes, and restrict visible modifications to relevant facial evidence. Otherwise, a plausible image may still be an unfaithful explanation of the model.

Localized diffusion counterfactuals face two dependent problems. First, localization must determine *where* the generator is permitted to act. Gradient saliency can identify pixels associated with a decision, but association does not show that modifying a highlighted semantic component changes that decision. Second, optimization must determine *how* to edit the chosen region. A weighted objective

$$L = w_T L_T + w_D L_D + w_I L_I + w_L L_L \qquad (1)$$

must use one set of weights across changing denoising geometry and incompatible states: reaching the target and preserving it after success. Strong preservation may block the target, whereas strong target guidance can move unrelated classifier outputs.

We connect the two decisions in one training-free framework. A disjoint discovery cohort supplies controlled masked diffusion interventions. Paired same-seed effects and source-ID-clustered confidence intervals verify which semantic regions influence the desired classifier output and form a frozen counterfactual influence graph. On a held-out source, this graph constrains Grad-CAM++ evidence to resolve the smallest verified semantic region set before generation. The resulting hard semantic mask defines what counts as an allowed change, and a corresponding soft mask localizes diffusion guidance and source blending.

Within that fixed support, we replace permanent weighting by a lexicographic local order. Target progress is primary; identity and outside-mask locality define the best attainable safety envelope; and non-target attribute drift is optimized only inside that envelope. Each reliable guided step solves a small trust-region problem in predicted-clean latent coordinates. A final restoration stage applies the same ordering to the actual post-blend latent.

Our contributions are:

- classifier-causal semantic mask discovery through paired same-seed region interventions and confidence-based influence-graph verification;
- source-specific minimal region resolution using a frozen graph and source evidence, without inspecting or reranking generated candidates;

- predicted-clean lexicographic trust-region guidance and preservation-aware final restoration confined to the resolved support.

Here, "causal" denotes an interventional effect inside a fixed explained classifier and generator. It does not identify biological, anatomical, social, or other real-world causes.

2 Related Work

Diffusion counterfactual explanations. Generative counterfactual methods use structured latent spaces to produce plausible alternatives [15, 8]. DiME and ACE combine classifier objectives with diffusion image priors [9, 10]; TIME and ECED extend diffusion counterfactual construction [11, 14]; and DiG-IN optimizes diffusion variables for model-analysis objectives [1]. A prior facial-counterfactual system combines face-part masks, diffusion generation, and adversarial analysis [20]. Our focus is the connection between intervention- verified localization and priority-constrained generation.

Saliency and mask-based localization. Grad-CAM and Grad-CAM++ localize classifier evidence through gradients [18, 5], but their maps remain observational explanations. Blended latent diffusion (BLD) confines synthesis by restoring a correspondingly noised source latent outside a spatial mask after each scheduler step [2]. MaskDiME derives dynamic classifier-gradient masks during denoising [7]. In contrast, we use Grad-CAM++ only to screen and resolve semantic candidates; causal verification comes from paired diffusion interventions, and the selected support is frozen before counterfactual generation.

Predicted-clean and multi-objective guidance. Latent diffusion samples in an autoencoder space [16], and DDIM supplies the scheduler used here [19]. Universal Guidance and diffusion posterior sampling demonstrate external measurements on predicted-clean estimates [3, 6]. Gradient- surgery methods modify conflicting multi-task directions [21], but a weighted or projected direction still leaves the priority policy implicit. We express that policy as sequential local subproblems: attain target progress, minimize unavoidable safety violation, then reduce collateral classifier drift.

3 Methodology

3.1 Preliminaries

Let $x_i \in [0,1]^{3\times H\times W}$ be a source image, $f_j(x)$ the logit of attribute j, T the explained target, and $y_T \in \{0,1\}$ its desired value. The desired probability is $p^\star(x) = \sigma((2y_T-1)f_T(x))$. Let $\mathcal{R}$ denote semantic component regions and $R \subseteq \mathcal{R}$ an allowed region set.

For fixed generation seed s, define the paired intervention effect

$$\Delta_{i,s,R} = p^\star\left(x_{i,s}^{\mathrm{do}(R)}\right) - p^\star(x_i). \tag{2}$$

The operation $\mathrm{do}(R)$ permits the generator to modify exactly R while source, prompt, seed, model, scheduler, and generation settings are held fixed. This is a classifier–generator intervention, not a claim that the semantic region causes an attribute in people or in the physical world.

3.2 Overall Framework

Figure 1 summarizes the complete method. A discovery cohort is processed once to screen semantic candidates, estimate controlled intervention effects, and freeze a reusable influence graph. For each new source, the graph and source Grad-CAM++ evidence select one minimal verified semantic region set. Selection occurs before synthesis and yields both a hard semantic mask and a soft generation mask. The former defines the allowed explanatory support; the latter provides smooth latent support for gradients and BLD blending. Predicted-clean lexicographic guidance, DDIM sampling, source blending, and final latent restoration then generate one audited counterfactual without expanding the chosen region.

3.3 Classifier-Causal Mask Discovery

Candidate screening. For each discovery source, Grad-CAM++ is intersected with semantic component masks. Captured attribution mass and within-component density rank plausible regions. This stage reduces the intervention search space but supplies no causal verification: high attribution is treated only as a proposal.

Controlled region interventions. We generate singleton and selected joint region interventions under the paired conditions of Eq. (2). The mask is the only allowed source of spatial variation; prompt, seed, model, scheduler, and inference settings remain identical. Post-generation attack is disabled so the measured effect belongs to localized generation rather than a subsequent optimizer. Alongside desired-probability change, the discovery record stores semantic mask cost, strict outside change, identity, non-target drift, and joint-region synergy.

Influence-graph verification. We aggregate repeated observations with a source-ID-clustered bootstrap, so multiple seeds of the same source are not treated as independent images. A target-to-region edge is verified only when the singleton region has positive mean effect, the lower confidence bound exceeds zero, and the frozen minimum sample count is met. Joint interventions estimate whether a union supplies additional effect beyond its singleton members. The verified edges,

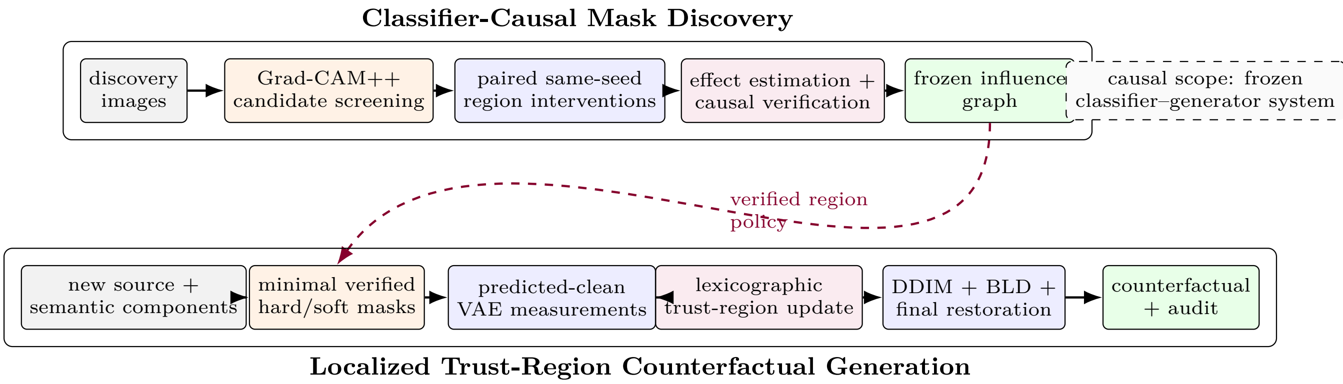

Figure 1: Overall framework. Controlled region interventions turn saliency-screened semantic components into a frozen influence graph. For a new source, the graph determines where editing is permitted; adaptive trust-region CCI determines how to edit within that fixed support. The graph is global and reusable, whereas the mask, counterfactual, and audit are source-specific.

costs, confidence intervals, and synergies form a frozen classifier-specific counterfactual influence graph.

Source-specific resolution. For a held-out source, we intersect its Grad-CAM++ map with its semantic masks and consider only regions admitted by the frozen graph. Among verified unions, we choose the smallest-cost set reaching the predeclared saliency-coverage requirement. Resolution can use only the source image, frozen graph, source Grad-CAM++ map, and semantic masks. It cannot inspect generated outputs, rerank candidates, retry a failed explanation, or expand the region after observing the counterfactual. This separation prevents output-dependent mask selection from masquerading as discovery.

3.4 Localized Trust-Region CCI

Localized objective. For required desired probability p_{req}, define

$$\kappa = \log \frac{p_{\text{req}}}{1 - p_{\text{req}}}, \qquad c_T(z) = \kappa - (2y_T - 1) f_T(x(z)). \quad (3)$$

The target requirement is met when $c_T(z) \leq 0$. The resolved binary semantic mask m_s states where change is allowed to count, while a dilated, feathered generation mask $m_g \in [0, 1]$ supplies smooth latent support and source blending. Non-target preservation includes all other classifier outputs:

$$D_{\text{NT}}(z) = \frac{1}{K-1} \sum_{j \neq T} \text{Huber}_\delta\big(p_j(x(z)) - p_j(x_i)\big). \quad (4)$$

Identity and locality define safety residuals

$$c_I(z) = 1 - \cos(e(x(z)), e(x_i)) - \varepsilon_I, \quad (5)$$

$$c_L(z) = \text{mean}\big((1 - m_g)|x(z) - x_i|\big) - \varepsilon_L. \quad (6)$$

Identity uses a frozen FaceNet representation trained on VGGFace2 [17, 4].

Predicted-clean measurements. At denoising step t, let z_t be the current latent, $\epsilon_\theta(z_t, t)$ the detached U-Net prediction, and $\bar{\alpha}_t$ the scheduler cumulative alpha. For epsilon prediction,

$$\hat{z}_0 = \frac{z_t - \sqrt{1 - \bar{\alpha}_t}\, \epsilon_\theta(z_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (7)$$

$$\hat{x}_0 = D(\hat{z}_0 / s_{\text{VAE}}), \qquad s_{\text{VAE}} = 0.18215. \quad (8)$$

The target, drift, identity, and locality models evaluate $\hat{x}_0$. Each image-space gradient passes through the fixed VAE decoder,

$$g = J_D(\hat{z}_0)^\top \nabla_x L, \quad (9)$$

but not through the U-Net. The decoder is therefore a differentiable measurement map, not another diffusion transition.

Lexicographic trust-region update. Let $g_T = \nabla c_T$, $g_D = \nabla D_{\text{NT}}$, $g_I = \nabla c_I$, and $g_L = \nabla c_L$ with respect to $\hat{z}_0$. Generation support is applied before norms, inner products, or optimization. If the target is unmet, requested local progress is

$$q_t = \min(\max(c_T, 0),\ \chi \Delta_t \|g_T\|_2), \quad (10)$$

where Δ_t is the clean-latent trust radius and $0 < \chi < 1$. The first subproblem finds the smallest unavoidable worst safety residual:

$$\tau^\star = \min_{d,\tau} \quad \tau \quad (11)$$

$$\text{s.t.} \quad g_T^\top d \leq -q_t, \quad c_I + g_I^\top d \leq \tau, \quad c_L + g_L^\top d \leq \tau, \quad \|d\|_2 \leq \Delta_t. \quad (12)$$

Inside that best envelope, the second subproblem minimizes collateral drift:

$$d^\star = \arg\min_{d} \quad g_D^\top d + \frac{1}{2\eta_t}\|d\|_2^2 \tag{13}$$

$$\begin{aligned} \text{s.t.} \quad & g_T^\top d \le -q_t, \\ & c_I + g_I^\top d \le \tau^\star + \zeta, \\ & c_L + g_L^\top d \le \tau^\star + \zeta, \\ & \|d\|_2 \le \Delta_t. \end{aligned} \tag{14}$$

Once the target is met, its progress constraint becomes the guard $c_T + g_T^\top d \le 0$. Deterministic bisection reduces an infeasible request; an unreliable target gradient skips the update. Because only four masked gradients participate, the subproblems are solved by active-set enumeration in their span using a small float64 Gram matrix. The radius adapts from achieved target progress: it shrinks below one-quarter of the accepted request and may grow when more than three-quarters is realized at the boundary.

The accepted clean displacement maps exactly to epsilon-prediction space:

$$\Delta\epsilon_t = -\sqrt{\frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}}\, d^\star. \tag{15}$$

After the scheduler step, BLD restores source content outside m_g:

$$z_{t-1} \leftarrow m_g \odot z_{t-1} + (1-m_g) \odot z_{t-1}^{\mathrm{src}}. \tag{16}$$

Fixed-weight CCI uses the same predicted-clean measurements, semantic support, clean-coordinate trust budget, constraints, and restoration budget, but applies a permanent nominal gradient composition. Adaptive trust-region CCI instead solves Eqs. (12)–(14) at the current iterate, so its effective coefficients follow current residuals and gradient geometry.

Preservation-aware final restoration. The final post-BLD latent may remain below the target threshold because a local step is attenuated by the scheduler or source blend. We therefore run at most 12 restoration iterations on that actual latent. Each iteration recomputes the two lexicographic subproblems through the fixed decoder and evaluators and tests fractions $\{1, 1/2, 1/4, 1/8\}$ of the proposed step. Before target success, acceptance requires strictly improved desired probability without worsening the best identity/locality envelope beyond tolerance. After target and safety success, it must maintain both and strictly decrease non-target drift. Every accepted step remains inside a cumulative clean-latent radius; failure of every fraction stops restoration. This stage uses no new U-Net prediction or scheduler transition. The saved uint8 image is evaluated again because VAE decoding and quantization can move a boundary-adjacent result.

4 Experimentation

4.1 Evaluation Protocols and Dataset

Completed localized-generation study. We evaluate smile removal (`Smiling` $1 \to 0$) on CelebAMask-HQ [13, 12]. Two random cohorts contain 100 completed sources each; all 200 source IDs are unique across cohorts and paired across methods. To isolate localized generation, this study uses the same fixed perioral union of mouth, upper-lip, and lower-lip components for all methods, with no automatic mask discovery, candidate reranking, or post-generation attack.

All methods use the local Stable Diffusion 2.1 base checkpoint, DDIM, 512^2 resolution, 35 scheduler steps, classifier-free guidance 5.0, blend start fraction 0.25, seed 42, float32 MPS execution, and batch size one. The desired in-loop probability is 0.8. Adaptive trust-region CCI uses clean radii 0.15/0.01/0.30, target progress fraction 0.9, reliability alpha threshold 0.10, Huber transition 0.02, at most 12 restoration iterations, and cumulative restoration radius 1.8. We compare BLD without CCI, Fixed-weight CCI, and Adaptive trust-region CCI (Ours).

Planned mask-discovery study. A discovery cohort will build the frozen influence graph, and a disjoint held-out cohort will evaluate source-specific masks. The study will report verified regions, paired intervention effects and source-ID-clustered confidence intervals, selected mask area, held-out target success, identity, outside-mask locality, and non-target drift. The completed 200-image study cannot supply these results because it uses a fixed mask.

Planned end-to-end attacked study. The full protocol compares BLD and Adaptive trust-region CCI on 300 paired images with mouth-only support, then on the same 300 IDs with mouth, upper-lip, and lower-lip support. It includes the localized smooth boundary attack. The same frozen multi-label classifier guides the attack and CCI and supplies all classifier-dependent metrics by design.

Metrics. Target@0.8 is the fraction reaching desired probability 0.8 under the explained classifier; it is stricter than flip rate (FR) at the 0.5 decision boundary. Identity is source/output FaceNet cosine. Outside MAE is strict outside-mask RGB error on $[0, 255]$. Non-target (NT) drift is the mean absolute probability change over all 39 non-target attributes. Runtime is median wall time per image. The attacked study additionally uses Fréchet inception distance (FID), spatial FID (sFID), face verification accuracy (FVA), feature similarity (FS), mean number of altered concepts (MNAC), collateral distance (CD), counterfactual output score (COUT), and FR. COUT, FR, MNAC, and CD all use the same frozen multi-label classifier that guides CCI; this is explained-model evaluation rather than independent-oracle validation.

4.2 Results and Analysis

Localized generation. Table 1 reports the completed clean comparison. Adaptive trust-region CCI lowers NT drift by 11.8% relative to Fixed-weight CCI (0.0635 versus 0.0720) while its Target@0.8 rate is one percentage point lower (37.5% versus 38.5%). Mean identity differs by 0.0034 and outside MAE by 0.0057 on $[0, 255]$. This supports reduced collateral classifier movement at a closely matched, but not identical, target operating point; it does not establish statistical superiority because paired confidence intervals are unavailable.

Both guided methods improve Target@0.8 and mean identity over BLD in this cohort, but BLD is substantially cheaper. Median runtime is 85.9 s for adaptive CCI, 81.8 s for fixed-weight CCI, and 20.4 s for BLD. Repeated VAE decodes, evaluator forwards/backwards, and restoration line-search evaluations dominate this overhead; the small active-set solve does not.

Restoration mechanism. For source 26811, restoration accepts 8 steps and raises desired probability from 0.1046 to 0.7946 while identity increases from 0.8348 to 0.8704 (Table 2). NT drift also rises, from 0.0327 to 0.0600, exposing the cost of recovering a severely missed target. This is a mechanism illustration, not population evidence.

The final image differs from the no-restoration output by pixel MAE 0.000810, maximum normalized change 0.2941, and 4.04% changed pixels. These values validate that the accepted restoration steps alter the saved image; they do not estimate an average improvement.

Pending classifier-causal mask discovery. Table 3 reserves the disjoint discovery and held-out results. No values are inferred from the fixed-mask localized-generation study.

Pending attacked evaluation. Tables 4 and 5 remain empty until their complete paired cohorts are available; partial results are not inserted.

Scope and limitations. The completed evidence covers one facial attribute, one dataset, one diffusion backbone, a fixed perioral policy, and deterministic seed-42 generation. Its statistics are descriptive, and the fixed/adaptive target operating points are close rather than identical. The same classifier guides generation and supplies classifier-dependent metrics. This is appropriate for explaining that model, but may reward model-specific features and does not establish agreement with another classifier or with people. Independent transfer and human evaluation remain future validation.

The local linearized subproblems provide an auditable priority rule, not a global convergence guarantee for a changing non-convex diffusion trajectory. The meaning of locality is bounded by segmentation quality, and repeated VAE and evaluator differentiation makes adaptive CCI slower than BLD. Finally, face images and identity embeddings are biometric data. Use should be limited to consented model auditing with suitable data protection, not sensitive-trait inference or identity-document modification.

5 Conclusions

We presented one framework in which intervention-verified localization determines where a counterfactual may change and adaptive lexicographic trust-region guidance determines how it changes within that fixed support. Paired same-seed interventions and source-ID-clustered confidence intervals turn saliency-screened semantic components into a reusable influence graph; held-out source evidence resolves a minimal mask without output inspection. Predicted-clean measurements then place heterogeneous evaluators on an image estimate, clean-coordinate trust regions enforce target-first priorities, and final restoration applies the same priorities to the actual post-BLD latent.

On 200 completed fixed-mask smile-removal sources, adaptive trust-region CCI reduces non-target drift by 11.8% relative to Fixed-weight CCI at a one-point lower Target@0.8 rate, with close identity and outside-mask error but higher runtime. Automatic discovery and attacked 300-image results remain pending and are not implied by this comparison. More broadly, every output is generated evidence about a frozen classifier– generator system; it must not be presented as a plausible real-world causal statement.

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Table 1: Completed clean smile-removal comparison over 200 unique paired sources using the same fixed perioral mask. Target@0.8 is in percent and Outside is strict outside-mask MAE on $[0, 255]$. Lower NT drift, Outside, and Time are better; higher Target@0.8 and Identity are better.

Method	Target@0.8 ↑	Identity ↑	Outside ↓	NT drift ↓	Time (s) ↓
BLD	17.5	0.8757	4.340	0.0477	20.4
Fixed-weight CCI	38.5	0.9255	4.334	0.0720	81.8
Adaptive trust-region CCI (Ours)	37.5	0.9221	4.339	0.0635	85.9

Table 2: Final-restoration illustration for source 26811.

Metric	Without	With
Desired probability	0.1046	0.7946
Identity cosine	0.8348	0.8704
NT drift	0.0327	0.0600
Runtime (s)	53.5	85.1

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Table 3: Planned automatic mask-discovery evaluation. The completed 200-image study uses a fixed perioral mask and therefore does not estimate these fields.

Target	Verified region(s)	Effect	95% CI	Mask area	Held-out success
Smile removal	Pending disjoint discovery/held-out run				

Table 4: End-to-end attacked evaluation with mouth-only support on 300 paired images.

Method	FID ↓	sFID ↓	FVA ↑	FS ↑	MNAC ↓	CD ↓	COUT ↑	FR (%) ↑
BLD	Pending complete 300-image paired run							
Adaptive trust-region CCI (Ours)	Pending complete 300-image paired run							

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Table 5: End-to-end attacked evaluation with mouth, upper-lip, and lower-lip support on the same 300 IDs.

Method	FID ↓	sFID ↓	FVA ↑	FS ↑	MNAC ↓	CD ↓	COUT ↑	FR (%) ↑
BLD	Pending complete 300-image paired run							
Adaptive trust-region CCI (Ours)	Pending complete 300-image paired run							